



MONASH University

MONASH UNIVERSITY

FACULTY OF INFORMATION TECHNOLOGY

FIT5032 Internet Applications Development

Assessed Lab 2

One-Way Data Binding, Computed Properties, and Vue Directives

Student Name	Wei Zhang
Student ID	36668699
Tutorial/Lab	Tuesday 2:00 pm
Project Name	wzhang-library

Tutor Name: Yujie Wang

3 July 2026

1 Lab Overview

This report presents the evidence completed for FIT5032 Assessed Lab 2. The lab focuses on one-way data binding in Vue.js 3, including importing JSON data, creating computed properties, rendering data with built-in directives, conditionally displaying content, dynamically binding attributes and styles, and managing the completed project through GitHub.

All assessed implementation work was completed individually. Peer interaction was limited to technical discussion and mutual support during the laboratory session.

2 Project Setup and JSON Data Import

2.1 Vue.js Project

A Vue.js 3 project named `wzhang-library` was created for the Week 2 laboratory activity. The project dependencies were installed using:

```
npm install
```

The development server was started using:

```
npm run dev
```

2.2 Activity 1: Importing the Provided Files

The files `authors.json` and `bookstores.json` were placed in:

```
src/assets/json
```

The provided component `JSON.vue` was placed in:

```
src/components
```

Both JSON files were imported into `JSON.vue` and used as the data source for the remaining activities.

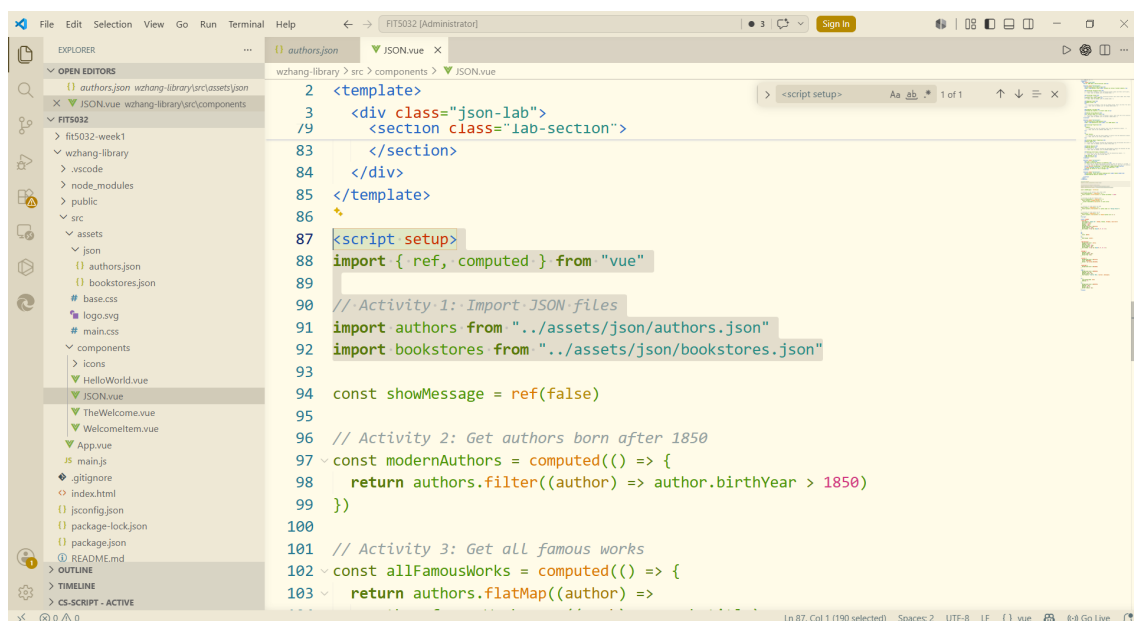


Figure 1: The project structure and JSON imports used for Activity 1.

Brief description: Figure 1 shows that the supplied JSON files and Vue component were placed in the required project directories and imported successfully.

3 eFolio Task 2.1: Computed Properties and Built-in Directives

3.1 Computed Property Activities

3.1.1 Activity 2: Authors Born After 1850

A computed property named `modernAuthors` was implemented using the JavaScript `filter()` method. It returns authors whose `birthYear` is greater than 1850.

3.1.2 Activity 3: All Famous Works

A computed property named `allFamousWorks` was implemented using `flatMap()` and `map()`. It combines the titles of famous works from all authors into one array.

3.1.3 Activity 4: Searching by Author Name

A computed property was used to locate an author by a specified name. For this activity, the search condition was set to `George Orwell`.

3.1.4 Activity 5: Searching by Author ID

A computed property was used to locate an author by a specified identifier. The selected author ID and the displayed result were verified in the running application.

3.2 List Rendering Activities

3.2.1 Activity 6: Author Names and Birth Years

The `v-for` directive was used to iterate through the authors array and display each author's name and birth year. A unique `:key` value was bound to each rendered list item.

3.2.2 Activity 7: Authors Born After 1850

The `v-for` directive was used to render the `modernAuthors` computed property created in Activity 2.

3.2.3 Activity 8: Famous Works

The `v-for` directive was used to render every title in the `allFamousWorks` computed property created in Activity 3.

3.3 Activities 9–12

Activities 9–12 used nested JSON data to display Jane Austen's famous works, bookstore company information, the total number of stores, physical and online store counts, opening hours, operating countries, and best-selling books. These activities demonstrated how to access nested objects and arrays and render their values in the Vue.js template.

3.4 Activity 13: Toggling Message Visibility

A Boolean reactive value named `showMessage` was used with `v-if` and `v-else`. A button click event toggles the value, causing the displayed message to change between the default prompt and the success message.

3.5 Complete Application Evidence

The following screenshots present the completed Vue.js application. Multiple screenshots are included so that the output remains clear and readable rather than being reduced to a single scaled-down image.

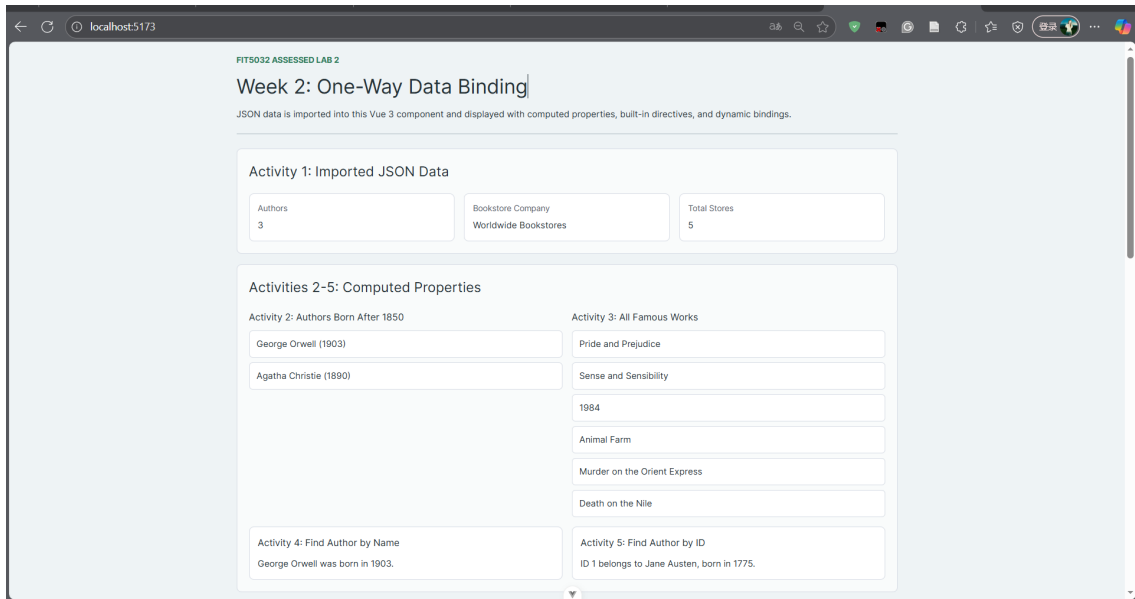


Figure 2: Upper section of the completed Vue.js application for Activities 1–13.

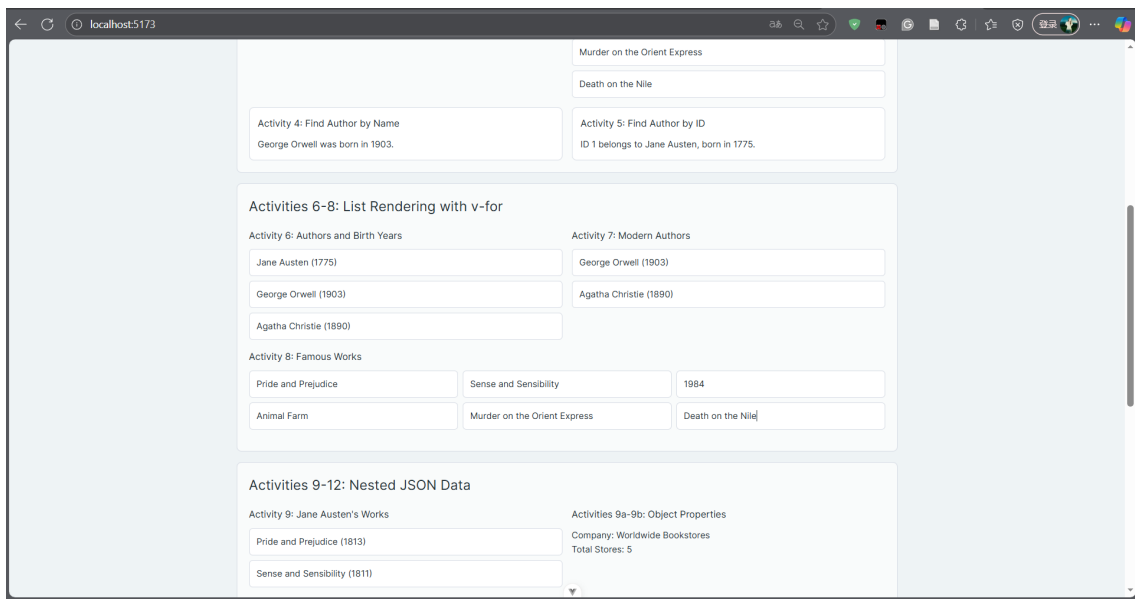


Figure 3: Middle section of the completed Vue.js application for Activities 1–13.

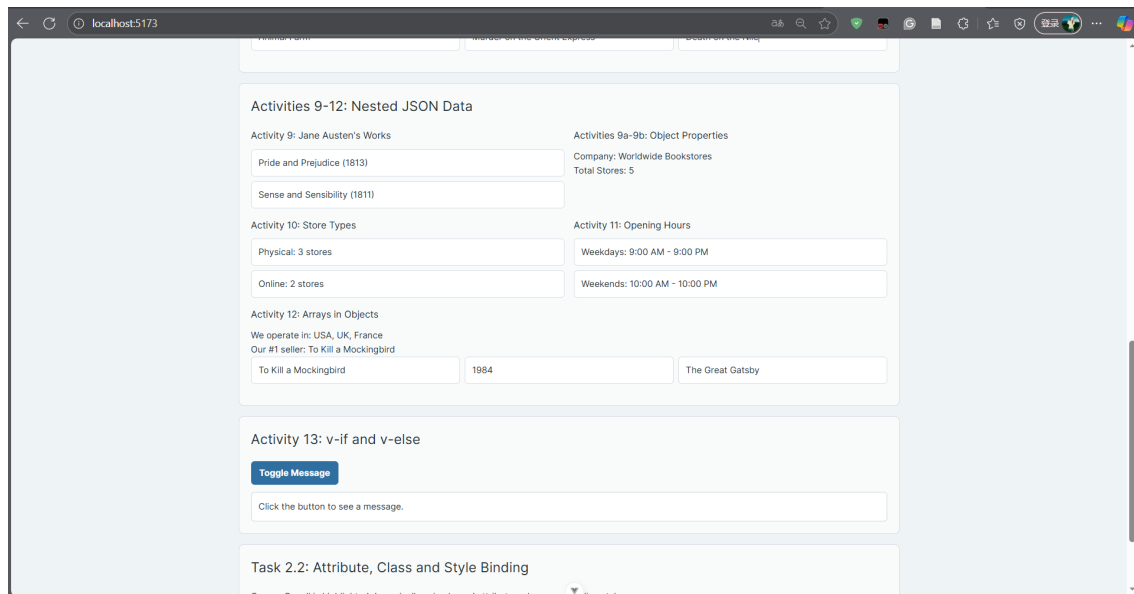


Figure 4: Lower section of the completed Vue.js application for Activities 1–13.

3.6 Message Toggle Evidence

Figure 5 shows the default message before the button is selected. Figure 6 shows the success message after the button is selected.

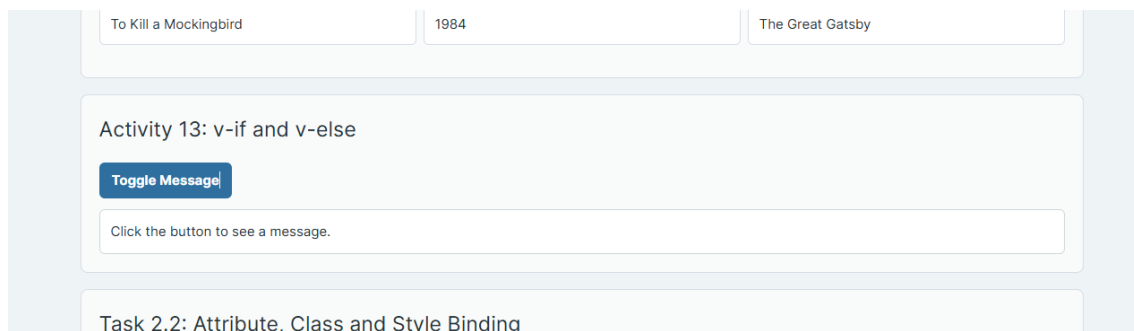


Figure 5: Activity 13 before the message is toggled.

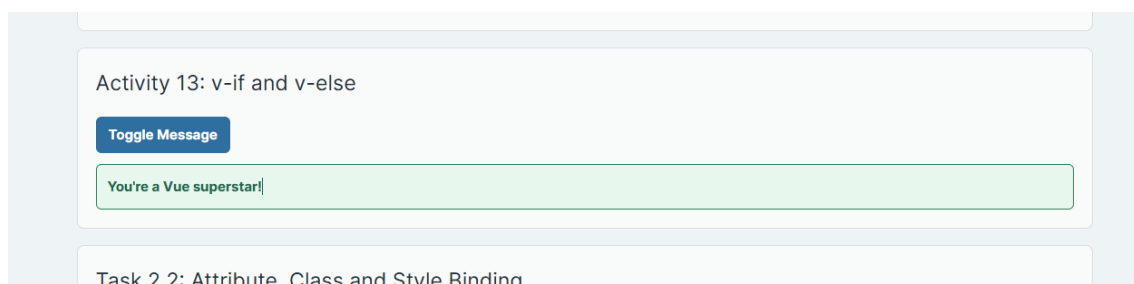


Figure 6: Activity 13 after the message is toggled.

4 eFolio Task 2.2: Attribute, Class, and Style Binding

4.1 George Orwell Highlight Functionality

Dynamic binding was used to identify the author George Orwell and apply a distinct visual style to the corresponding rendered item. The condition was based on the author data rather than a manually

formatted static HTML element.

Figure 7 shows the completed functionality in the browser. George Orwell is visually distinguishable from the other authors.

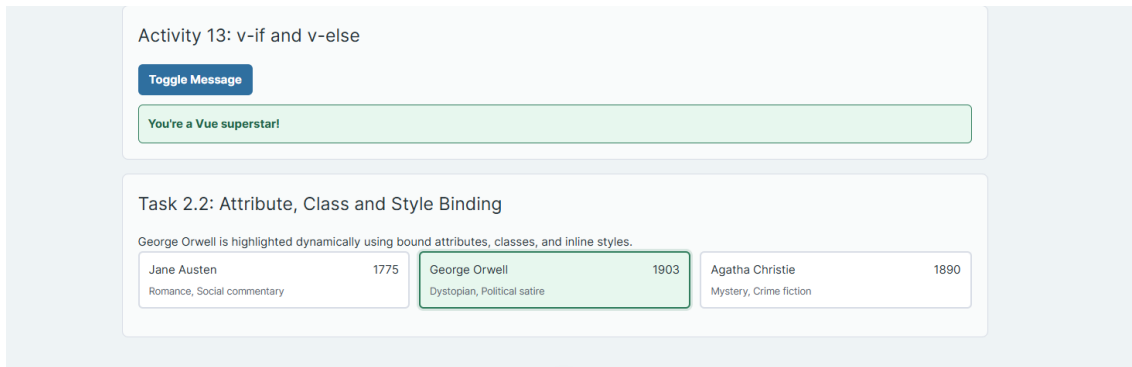


Figure 7: The running application with George Orwell dynamically highlighted.

4.2 Implementation Evidence

Figure 8 presents the Vue template binding and the related CSS or style definition used to implement the author highlight functionality.

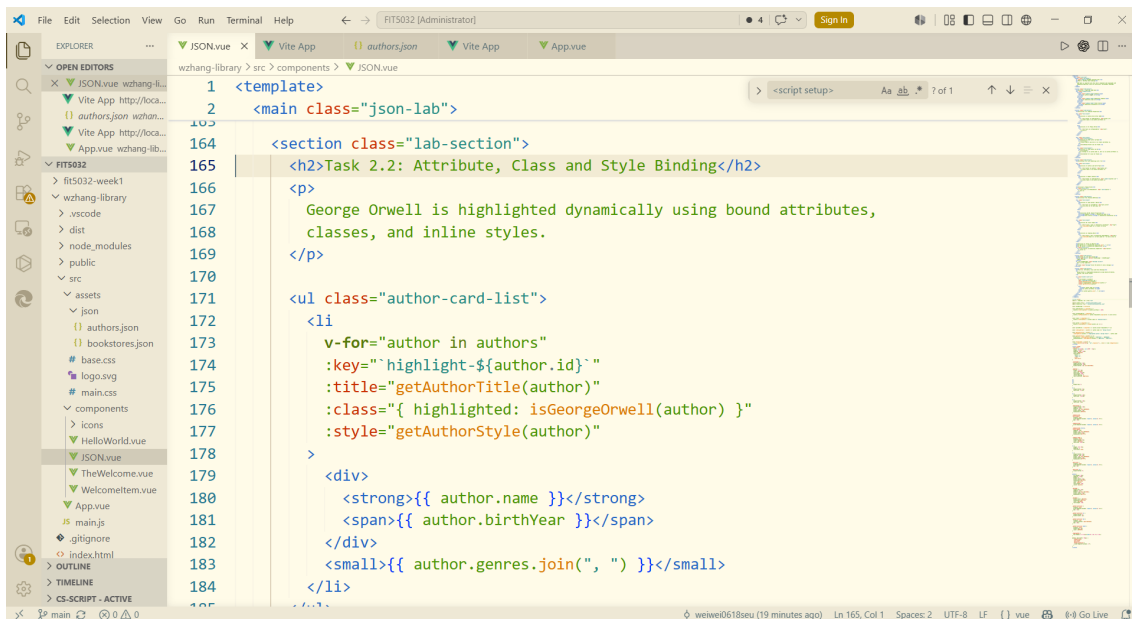


Figure 8: Source code implementing the George Orwell highlight functionality.

Brief description: The implementation uses Vue's dynamic binding syntax, such as `:class`, `:style`, or another appropriate `v-bind` expression, to apply the highlight only when the author's name matches George Orwell.

5 GitHub Version Control Evidence

The completed Vue.js project was committed and pushed to GitHub for version management and maintenance.

Project Name	wzhang-library
GitHub Repository	https://github.com/weiwei0618seu/wzhang-library

Figure 9 shows the GitHub repository history after the Lab 2 project update was pushed successfully.

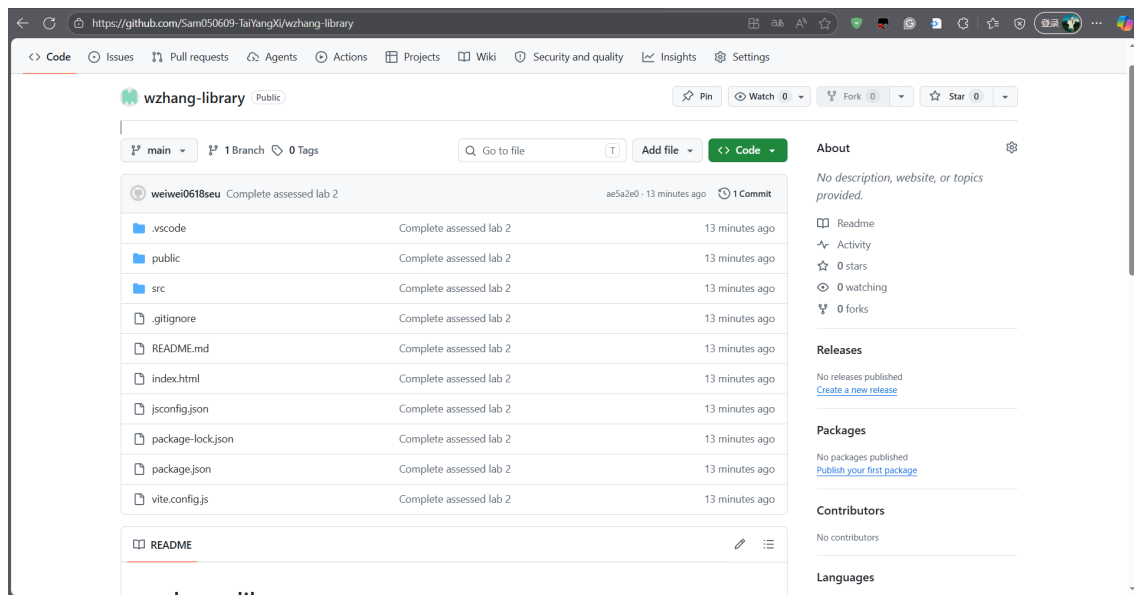


Figure 9: GitHub commit history showing the successful Lab 2 project update.

6 Evidence Checklist

Assessment Item	Evidence Included	Status
Task 2.1	Multiple readable screenshots showing the completed application and Activities 1–13	Complete
Activity 13	Screenshots showing the message before and after the button click	Complete
Task 2.2 Functionality	Screenshot showing George Orwell highlighted	Complete
Task 2.2 Code	Screenshot showing the dynamic binding implementation	Complete
GitHub History	Screenshot showing the successful project push and commit history	Complete

Table 1: Assessed Lab 2 evidence checklist.

7 Completion Summary

The required tasks for FIT5032 Assessed Lab 2 were completed successfully. JSON data was imported into the Vue.js project, computed properties were used to transform and search the data, and Vue built-in directives were used to render lists and conditionally display content. Dynamic attribute, class, or style binding was also implemented to highlight George Orwell. Finally, the completed project was committed and pushed to GitHub, and the required evidence has been included in this report.